

Home Assignment: Build a Mini E-Commerce App using React Context + useReducer

Apply what you've learned about useReducer and Context API by building a small e-commerce-style app.

The app will manage global state for authentication, theming, and cart functionality using clean, modular logic.

Objective

Create a React + TypeScript app with:

- Centralized state using Context + useReducer
 - Auth and theme switching
 - Cart logic with quantity, removal, and total price
 - Proper typing and architecture
-

Required Features

1. Cart Management

Use CartContext and cartReducer to manage cart state:

```
type CartItem = {  
  id: string;  
  title: string;  
  price: number;  
  quantity: number;  
};
```

You should be able:

- Add items to the cart
 - Remove items from the cart
 - Update quantity
 - Clear the entire cart
 - Show total price
-

2. Theme Management

Use ThemeContext with a reducer:

type Theme = "light" | "dark";

You should be able:

- Toggle between light/dark themes
 - Store preference in localStorage
 - Apply theme using className or Styled Components
-

3. Product Page

Create a product list from hardcoded data or fetch from <https://fakestoreapi.com/products>.

Each product should display:

- Title, image, price
 - "Add to Cart" button
-

4. Routing (React Router)

Use React Router for navigation:

- /products – product list (public)
 - /cart – cart (only if logged in)
 - /profile – username and logout option
-